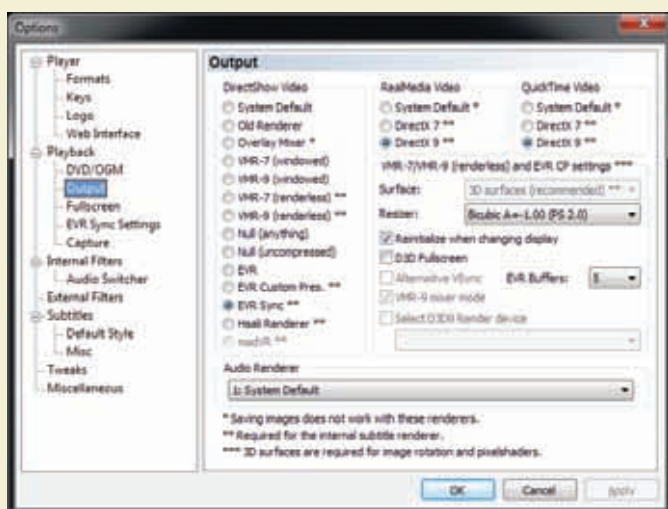




Choose the appropriate renderer to enhance video playback

Renderer. The benefits of using these renderers include the reduction or elimination of screen tearing, allowing cards to render video as textures which results in higher quality, and the facilitation of DXVA. Be warned though, if your PC is too much of a fossil, it won't be able to handle these renderers because you'll need hardware that's at least D3D9 capable.

Nearly every player has an option to select the renderer you want to use, so go ahead and choose. The exception being Windows Media Player, which defaults to EVR in Vista and 7 anyway. In MPC-HC, you'll find the renderers listed in the

Output tab, under the Playback section of the options.

Convert to RGB

YV12 is the default output colourspace for codecs, but using NV12 or RGB might result in better quality. NV12 because it's the colourspace used in DXVA, and is required for hardware de-interlacing. And RGB because it's generally more accurate and results in better quality subtitles – but taxes your CPU a tad more.

If you're indeed converting your videos to an RGB colour-space, and your CPU has power to spare, make sure to select the ITU-R BT.709 conversion method, check High quality

YV12 to RGB conversion and Dithering. The latter two options make the colour conversion happen in 11-bit or above, which otherwise happens in 8-bit and could result in a phenomenon known as colour banding, where pixels of a particular colour tend to "band" together, stand out from the rest of the image and look very low quality. Alternatively, you could also procure or encode your videos in a Hi10Profile, which ffdshow and CoreAVC now natively support.

Another thing that goes un-noticed is that while most videos, and indeed televisions, display images in a colour range of 16-255, Monitors expect a 0-255 signal from the PC. Setting the output levels to 0-255 will result in blacks that are slightly blacker, resulting in less washed out colours and more contrast. Be careful not to select the input range as 0-255 though, because that will result in doing the exact opposite – the loss of contrast and colour depth.

All these options can be found under Output and RGB Conversion in the ffdshow configuration window.

Upscale video

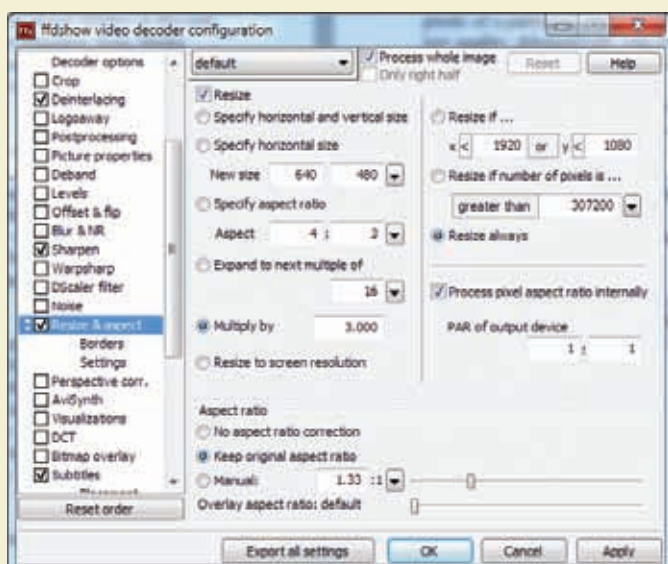
You can use ffdshow to upscale your video for drastic improve-

ments in quality, provided your source video is of a sufficient quality. This happens because using the decoder to resize the video is a better option than letting your GPU or Display stretch the image out to fit your screen. It also results in sharper subtitles of a much higher resolution than usual.

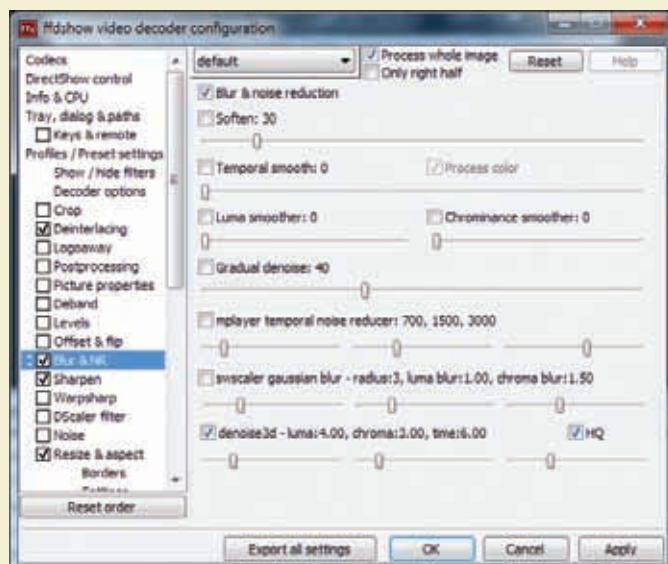
There are several resizing options to choose from. You can choose to resize the video to fit your screen resolution, or if you're hungry for even more quality use multiples to go as high as your processor allows you to. 2x or 3x is usually the sweet spot for a 1080p display, unless you're resizing a 1080p video and your processor can't handle it! Make sure to select "Process Whole Image" at the top of the config screen, though.

Denoise/sharpen/de-interlace

These are the other options you can use to extract some more quality out of your videos. Denoise 3D, which sits under the "Blur & NR" category in ffdshow, does just what you'd expect from a filter with that name – it seeks to remove noise from a video. It uses a lot of processing power though,



Correctly upscaling a video could result in noticeable quality gains



Enable denoise3d to reduce visible noise in the video